# Every-Iterate Quadratic Amplitudes for Finite Cyclic Rotations

Route-A structural certificate C161


**Abstract**

For the odd cyclic rotation $R_q(x) = x + 1$ with quadratic observable $ax^2 + bx$, we evaluate the complete Birkhoff amplitude at every iterate and for every source parameter. If $d = (an, q)$, the sum vanishes exactly when $d \nmid an(n-1) + bn$; otherwise its nonconstant branch has magnitude $d\sqrt{q/d}$ and an explicit Jacobi sign and completed-square phase. Over a prime source ring the zero level is $1 + (\Delta/p)$ in the quadratic branch; for $p \geq 5$ and $a = 1, b = 0$, $\Delta = n^2(1 - n^2)/3$. This all-parameter theorem replaces a Heisenberg candidate that failed its quotient-coordinate and local-equivalence proof gate. Exhaustive checks through odd $q \leq 31$ are regression sentinels, not the proof. A same-clock finite unitary realizes the amplitude as $\mathrm{Tr}(U^n K^{-n})$. No target trace law, arithmetic Euler data, or Hilbert–Pólya operator is claimed.



**Keywords:** cyclic dynamics; Birkhoff sum; quadratic Gauss sum; exact vanishing; discriminant; finite Koopman unitary.



中文摘要：对奇数阶循环旋转 $R_q(x) = x + 1$ 及二次观测量 $ax^2 + bx$，本文在任意参数与任意迭代下精确求出完整比尔科夫振幅。令 $d = (an, q)$；当且仅当 $d$ 不整除 $an(n-1) + bn$ 时振幅为零；否则非平凡分支的模长为 $d\sqrt{q/d}$，符号与相位分别由雅可比符号和 配方余数确定。素数阶情形的零水平数由判别式 $\Delta$ 控制；当 $p \geq 5$ 时，纯二次子族满足 $\Delta = n^2(1 - n^2)/3$。此前海森堡模型的商坐标及 2、5 进局部等价未完成严格证明，故按硬门槛更换模型。有限范围穷举仅用于回归验证；理论证明来自公因数约化和高斯和求值。本文不主张目标迹公式、欧拉数据或希尔伯特波利亚算子。



关键词：循环动力学；比尔科夫和；二次高斯和；精确消失；判别式；有限库普曼西算子。


## 1 All-iterate evaluation

Let $q$ be odd, let $n \geq 1$, let $R_q(x) = x + 1$ on $\mathbb{Z}/q\mathbb{Z}$, and $\phi_{a,b}(x) = ax^2 + bx$. Summing $1, j, j^2$ gives

$$\begin{gathered} S_n\phi(x) = A_n x^2 + B_n x + C_n \pmod q, \\ A_n = an, \quad B_n = an(n-1) + bn, \\ C_n = \frac{an(n-1)(2n-1)}{6} + \frac{bn(n-1)}{2}. \end{gathered} \tag{1}$$

The quotients in $C_n$ are integers before reduction. Define

$$G_{q,n}(a, b) = \sum_{x \bmod q} \exp(2\pi i S_n\phi(x)/q).$$

Put $d = (A_n, q)$ and $Q = q/d$. Translation $x \mapsto x + Q$ multiplies the sum by $\exp(2\pi i B_n/d)$. Hence $G_{q,n} = 0$ exactly when $d \nmid B_n$. If $d \mid B_n$ and $Q > 1$, write $A' = A_n/d$, $B' = B_n/d$, and

$$h \equiv -(4A')^{-1}(B')^2 \pmod Q, \qquad r \equiv C_n + dh \pmod q.$$

Reduction gives $d$ copies of a primitive odd quadratic sum, and completing the square gives

$$G_{q,n}(a, b) = d\left(\frac{A'}{Q}\right)\epsilon_Q\sqrt{Q}\,e^{2\pi i r/q}, \qquad \epsilon_Q = \begin{cases} 1, & Q \equiv 1 \pmod 4), \\ i, & Q \equiv 3 \pmod 4). \end{cases} \tag{2}$$

Here $(A'/Q)$ is the source-ring Jacobi symbol. For $\Gamma_k(A) = \sum_{x \bmod p^k} e^{2\pi i A x^2/p^k}$, split $x = y + p^{k-1}z$. The $z$-sum vanishes unless $p \mid y$ and gives

$$\Gamma_k(A) = p\,\Gamma_{k-2}(A) \quad (k \geq 2).$$

Together with $\Gamma_1(A) = (A/p)\epsilon_p\sqrt{p}$ and $\Gamma_0 = 1$, this proves every odd prime-power case. CRT combines the prime-power sums, and quadratic reciprocity reduces the accumulated signs to $(A'/Q)\epsilon_Q$. Thus (2) is an exact finite-sum identity. If $Q = 1$, the separate constant branch is $G_{q,n} = qe^{2\pi i C_n/q}$. Thus (2), including its vanishing gate, holds for every $q, a, b, n$ in the frozen family; finite sweeps are only regressions.

## 2 Zero levels and a concrete specialization

For an odd prime $p$, reduce $(A_n, B_n, C_n)$ modulo $p$. If $A_n \neq 0$, completion of the square yields

$$\#\{x : S_n\phi(x) = 0\} = 1 + \left(\frac{\Delta}{p}\right), \qquad \Delta = B_n^2 - 4A_nC_n. \tag{3}$$

If $A_n = 0 \neq B_n$ the count is one; if $A_n = B_n = 0$, it is $p$ or zero according as $C_n = 0$ or not. In the pure quadratic case $a = 1, b = 0$, $p \geq 5$, and $n \not\equiv 0 \pmod p$,

$$\Delta = \frac{n^2(1-n^2)}{3}, \qquad \#\{x : S_nx^2 = 0\} = 1 + \left(\frac{\Delta}{p}\right). \tag{4}$$

Thus $n \equiv \pm 1 \pmod p$ gives the single double root, whereas $n \equiv 0 \pmod p$ gives all $p$ roots.

## 3 Same-clock unitary and boundary

On $\mathcal{H}_q = \ell^2(\mathbb{Z}/q\mathbb{Z})$ let $(K_qf)(x) = f(x+1)$, let $M_\phi$ multiply by $e^{2\pi i\phi(x)/q}$, and put $U_\phi = M_\phi K_q$. Direct iteration gives the exact same-clock identity

$$G_{q,n}(a, b) = \mathrm{Tr}(U_\phi^n K_q^{-n}). \tag{5}$$

The compensating shift is essential; (5) is not replaced by $\mathrm{Tr}(U_\phi^n)$. Let $Pf(x) = f(-x)$, let $J$ be complex conjugation, set $g(x) = (a-b)x^2$, and let $D_g$ multiply by $e^{2\pi i g(x)/q}$. The antiunitary $\Theta = D_gPJ$ obeys

$$\Theta^2 = I, \qquad \Theta U_\phi \Theta^{-1} = U_\phi^{-1}. \tag{6}$$

Indeed $g$ is even and $g(x) - g(x-1) = \phi(-x) - \phi(x-1) = (a-b)(2x-1)$, exactly the multiplier identity obtained when $P$ reverses the shift. More explicitly,

$$(\Theta U_\phi \Theta f)(x) = e^{2\pi i\{g(x)-\phi(-x)-g(1-x)\}/q} f(x-1) = e^{-2\pi i\phi(x-1)/q} f(x-1) = U_\phi^{-1} f(x).$$

The strict tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_NATURAL_QUANTIZATION`). We claim no target trace/divisor/counting law, isolated stability determinant, arithmetic local/Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.

## Declarations

**Data availability.** All claim-bearing data and programs are packaged with this certificate; no external dataset is used. **Ethics.** No human participants, animals, or sensitive personal data are involved. **Author contributions/CRediT.** Anonymous technical certificate; Conceptualization, Formal analysis, Software, Validation, and Writing are recorded at package level. **Competing interests.** None known. **Funding.** No external funding is reported. **AI-use disclosure.** An AI coding assistant supported derivation planning, drafting, and code review. Packaged independent checks validate quantitative claims; the assistant was not treated as an external peer reviewer.